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Grade	10.00 out of 10.00 (100%)

Question 1

Correct

Mark 1.00 out of 1.00

When talking about choosing the right problem for your research, what is considered an "optimal" scientific problem?

- ☒ a. It depends on how you balance the difficulty and your interest. ✓
- ☐ b. A problem that does not require much work.
- ☐ c. A problem that no one has ever tried to solve, or succeeded in solving it.
- ☐ d. A problem that brings you a lot of new knowledge or has many practical uses.

The correct answer is: It depends on how you balance the difficulty and your interest.

Question 2

Correct

Mark 1.00 out of 1.00

How many citations were there to this paper (*How To Choose a Good Scientific Problem*) in 2023 according to Google Scholar?

- ☒ a. 17 citations ✓
- ☐ b. 16 citations
- ☐ c. 18 citations
- ☐ d. 19 citations

The correct answer is: 17 citations

Question 3

Correct

Mark 1.00 out of 1.00

During your research, your experimental results are not as good as you expected, and some of them even contradict current theories. What should you do next?

- ☒ a. Carefully investigate what went wrong: your experiment setup, your current materials, etc. If you still cannot figure out what was wrong, report this information to your supervisor to find out the solution. ✓
- ☐ b. There must be something wrong with the experiment. Just re-run the experiment until it fits with current theoretical ideas.
- ☐ c. Just report these numbers and pretend that you never heard of that theory.
- ☐ d. Stop this project, get depressed, and quit since this approach does not make any sense.

The correct answer is: Carefully investigate what went wrong: your experiment setup, your current materials, etc. If you still cannot figure out what was wrong, report this information to your supervisor to find out the solution.

Question 4

Correct

Mark 1.00 out of 1.00

In what month and year the paper (*How To Choose a Good Scientific Problem*) was published?

- ☐ a. 2009 Oct
- ☒ b. 2009 Sept ✓
- ☐ c. 2019 Sept
- ☐ d. 2019 Oct

The correct answer is: 2009 Sept

Question 5

Correct

Mark 1.00 out of 1.00

Which of the following practices is good during your problem-finding phase?

- ☐ a. Choose the biggest problem that the world is facing, even though your current infrastructure (lab equipment, computers, GPUs, etc.) cannot sufficiently handle that.
- ☒ b. Take time to think about the problem; carefully read, discuss, and plan what to do next if you pick that problem. ✓
- ☐ c. Start working as soon as possible on the first problem that came to your mind.
- ☐ d. Wait for your professor/supervisor recommendations. Their recommendations are always the best.

The correct answer is: Take time to think about the problem; carefully read, discuss, and plan what to do next if you pick that problem.

Question 6

Correct

Mark 1.00 out of 1.00

There is a 3 month rule for new students or postdocs. What is **NOT** correct about this rule?

- ☐ a. The new student or postdoc reads, discusses and plans his/her research. He/she focuses on being rather than doing.
- ☐ b. Do not commit to a problem before we spend 3 months.
- ☒ c. It is the period that the new student or postdoc reads, discusses, plans, carries out research and submit papers. ✓
- ☐ d. Spending weeks in choosing a good problem can save months or years later on.

The correct answer is: It is the period that the new student or postdoc reads, discusses, plans, carries out research and submit papers.

Question 7

Correct

Mark 1.00 out of 1.00

When you are looking for papers about your research topic, which of these is **NOT** a good source to use?

- ☒ a. A technical post shared by a friend on Threads. ✓
- ☐ b. Publicly available master's or PhD theses.
- ☐ c. Papers published in the proceedings of a CORE-ranked conference.
- ☐ d. Peer-reviewed journals that are listed in Scopus and ISI, and are in the Q1 or Q2 quartile.

The correct answer is: A technical post shared by a friend on Threads.

Question 8

Correct

Mark 1.00 out of 1.00

By which **publisher** was the paper *How To Choose a Good Scientific Problem* published?

*Note that the **publisher** is different from **the journal**.*

- ☐ a. Google Scholar
- ☒ b. Elsevier ✓
- ☐ c. Journal of Molecular Cell
- ☐ d. IEEE Xplore

The correct answer is: Elsevier

Question 9

Correct

Mark 1.00 out of 1.00

Our research proposal says we start at A and try to reach B (the established goal). When it is not reached, what should we do according to the nurturing scheme of research?

- ☒ a. When we enter the phase, namely, “the cloud”, we should sense a new problem in the materials at hand and move forward with the new goal C if it is more interesting and feasible. ✓
- ☐ b. We should stop the research and find a new research problem as our lab requires us to reach B.
- ☐ c. We should change our research area as the goal cannot be reached in this area.
- ☐ d. The objective scheme of research states we should proceed to B which is the shortest path. There is nothing else we can do.

The correct answer is: When we enter the phase, namely, “the cloud”, we should sense a new problem in the materials at hand and move forward with the new goal C if it is more interesting and feasible.

Question 10

Correct

Mark 1.00 out of 1.00

After a while, you have chosen a topic and after a quick search on Google Scholar, there are hundreds of papers related to the topic. What should you do next?

- ☒ a. Select a few of them, then read briefly over the papers and narrow down the total number of quality papers. You need to understand the background, the problem statements, approaches, and solutions to have an overview of the problem. ✓
- ☐ b. Read all of them to understand every little detail of each paper.
- ☐ c. Just read the latest paper to know what is the state-of-the-art (SOTA). We just need to focus on the latest, those outdated papers are not important.
- ☐ d. Give up since there are too many things that you've missed, there's no way you can keep up with the current research.

The correct answer is: Select a few of them, then read briefly over the papers and narrow down the total number of quality papers. You need to understand the background, the problem statements, approaches, and solutions to have an overview of the problem.